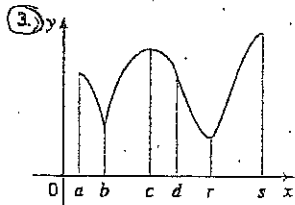


WTW158 Unit 4:1 Maximum and minimum values

Exercises 4.1 p 283

3-4 For each of the numbers a, b, c, d, r , and s , state whether the function whose graph is shown has an absolute maximum or minimum, a local maximum or minimum, or neither a maximum nor a minimum.



c - local max

d - neither local/absolute max or min

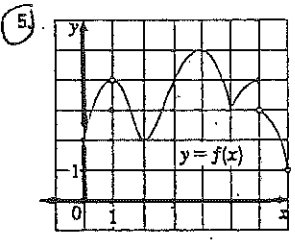
r - absolute and local min

s - absolute max

a - neither local/absolute max or local/absolute min

b - local min

5-6 Use the graph to state the absolute and local maximum and minimum values of the function.



• absolute max of 5 at $x=4$

• no absolute min

• local maximum of 5 at $x=4$

• local maximum of 4 at $x=6$

• local minimum of 2 at $x=2$

• local minimum of 3 at $x=1$ and $x=5$

29-44 Find the critical numbers of the function.

31. $f(x) = 2x^3 - 3x^2 - 36x$

33. $g(t) = t^4 + t^3 + t^2 + 1$

39. $F(x) = x^{4/5}(x-4)^2$

41. $f(\theta) = 2 \cos \theta + \sin^2 \theta$

31. Domain of f is $\mathbb{R}$

f is continuous and differentiable on $\mathbb{R}$

$$f'(x) = 0 \Rightarrow 6x^2 - 6x - 36 = 0$$

$$\Rightarrow 6(x^2 - x - 6) = 6(x-3)(x+2) = 0$$

$$\Rightarrow x = 3 \text{ or } x = -2$$

Critical numbers 3 or -2

33. g is continuous and differentiable on its domain, $\mathbb{R}$

$$g'(t) = 0 \Rightarrow 4t^3 + 3t^2 + 2t = t(4t^2 + 3t + 2) = 0$$

$$\Rightarrow t = 0 \text{ as } 4t^2 + 3t + 2 > 0$$

The only critical number is 0

(39) $F(x) = x^{4/5} (x-4)^2$

F is continuous on its domain $\mathbb{R}$.

$$F'(x) = 0 \Rightarrow \frac{4}{5} x^{-1/5} (x-4)^2 + x^{4/5} 2(x-4) = 0$$

$$\Rightarrow (x-4) \left[\frac{4}{5} x^{-1/5} (x-4) + 2x^{4/5} \right] = (x-4) \left[\frac{4}{5} x^{4/5} - \frac{16}{5} x^{-1/5} + 2x^{4/5} \right] = 0$$

$$\Rightarrow (x-4) \left(\frac{14}{5} x^{4/5} - \frac{16}{5} x^{-1/5} \right) = 0 \Rightarrow (x-4) \frac{1}{5} x^{-1/5} (14x - 16)$$

$$\Rightarrow x=4 \text{ or } 14x - 16 = 0$$

$$\Rightarrow x=4 \text{ or } x = \frac{16}{14} = \frac{8}{7}$$

The critical numbers are 4 , $\frac{8}{7}$ and 0 (if not defined at 0)

* (41) $f(\theta) = 2\cos\theta + \sin^2\theta$

f is continuous and differentiable on its domain $\mathbb{R}$.

$$f'(\theta) = 0 \Rightarrow -2\sin\theta + 2\sin\theta\cos\theta = 0$$

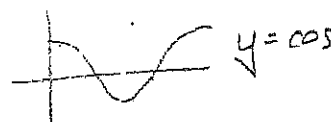
$$\Rightarrow 2\sin\theta(-1 + \cos\theta) = 0$$

$$\Rightarrow \sin\theta = 0 \text{ or } \cos\theta = 1$$

$$\Rightarrow \theta = k\pi \text{ or } \theta = 0 + 2k\pi$$

$$\Rightarrow \theta = 2k\pi$$

critical numbers are $2k\pi$, $k \in \mathbb{Z}$



Find the absolute maximum and absolute minimum values of f on the given interval

(49) $f(x) = 2x^3 - 3x^2 - 12x + 1$ on $[-2, 3]$

As f is continuous on a closed interval, use the Closed interval method.

$$f'(x) = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x-2)(x+1)$$

$$f'(x) = 0 \Rightarrow x = -1 \text{ or } x = 2$$

Critical numbers are -1 and 2

x	-2	-1	2	3
$f(x)$	-3	8	-19	-8

f has an absolute maximum of 8 at $x = -1$

f has an absolute minimum of -19 at $x = 2$

(57) $f(t) = 2\cos t + \sin 2t$ on $[0, \frac{\pi}{2}]$

Use the close interval method as f is continuous on $[0, \frac{\pi}{2}]$

$$\begin{aligned} f'(t) &= 2(-\sin t) + 2\cos 2t = -2\sin t + 2(1 - 2\sin^2 t) \\ &= -2\sin t + 2 - 4\sin^2 t = -2[2\sin^2 t + \sin t - 1] \\ &= -2(2\sin t - 1)(\sin t + 1) \end{aligned}$$

$$f'(t) = 0 \Rightarrow \sin t = \frac{1}{2} \text{ or } \sin t = -1$$

$$\Rightarrow t = \frac{\pi}{6}$$

no solution in interval $[0, \frac{\pi}{2}]$

The only critical number is $\frac{\pi}{6}$

t	0	$\frac{\pi}{6}$	$\frac{\pi}{2}$
$f(t)$	2	$\frac{3\sqrt{3}}{2}$	0

f has an absolute maximum of $\frac{3\sqrt{3}}{2}$ at $t = \frac{\pi}{6}$

f has an absolute minimum of 0 at $t = \frac{\pi}{2}$

WTW 158 Unit 4.1 Exercises 4.1 p 284

60) $f(x) = x e^{x/2}, [-3, 1]$

$f(x)$ is continuous on the given interval. We may therefore use the closed interval method

$$f'(x) = e^{x/2} + x e^{x/2} \cdot \frac{1}{2} = e^{x/2} \left(1 + \frac{1}{2}x\right)$$

$f'(x)$ exists everywhere and $f'(x) = 0$ when $x = -2$

We have to investigate $f(-3)$, $f(1)$ and $f(-2)$

$$f(-3) = -3 e^{-3/2}$$

$$f(-2) = -2 e^{-1}$$

$$f(1) = e^{1/2}$$

The absolute maximum is $f(1) = e^{1/2}$
With your calculator you will find
that the abs min is $f(-2) = -\frac{2}{e}$

* 61) $f(x) = \ln(x^2 + x + 1)$ on $[-1, 1]$

Use the close interval method as f is continuous on $[-1, 1]$

$$f'(x) = \frac{1}{x^2 + x + 1} \times 2x + 1 = \frac{2x + 1}{x^2 + x + 1}$$

$f'(x) = 0 \Rightarrow x = -\frac{1}{2}$. This is the only critical number

x	-1	$-\frac{1}{2}$	1
$f(x)$	0	$\ln \frac{3}{4}$	$\ln 3$

clearly
 $\ln \frac{3}{4} < 0 < \ln 3$

f has an absolute maximum at $\ln 3$ at $x = 1$

f has an absolute minimum at $\ln \frac{3}{4}$ at $x = -\frac{1}{2}$

WTW158 Unit 4.1 Exercises 4.1 p280

* 76 Show that 5 is a critical number of the function

$$g(x) = 2 + (x - 5)^3$$

but g does not have a local extreme value at 5.

$$g'(x) = 3(x - 5)^2$$

$$g'(x) = 0 \Rightarrow x = 5$$

The only critical number is 5

The function has no local extremes because it is

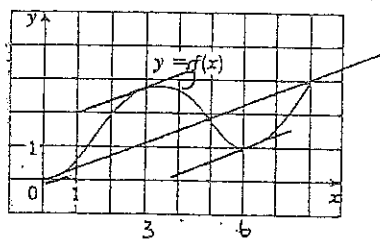
always increasing ($g'(x) > 0$ for $x \neq 5$) on any open

around $x = 5$, there are always function values less

than 2 and greater than 2 in that interval

WTW 158 Unit 4.2 Exercises 4.1 p 291

③ Use the graph of g to estimate the values of c that satisfy the conclusion of the Mean Value Theorem for the interval $[0, 8]$. : $C=3$ and $C=6$



The slopes of the 3 tangent lines at $x=3$ and $x=6$ and $x=0$ are equal to the slope of the secant line

a) g satisfies the hypotheses of the Mean Value theorem because g is (a) continuous on the closed interval $[0, 8]$ and (b) differentiable on the open interval $(0, 8)$

b) $g'(c) = \frac{g(8) - g(0)}{8 - 0} = \frac{4 - 1}{8} = \frac{3}{8}$. It appears that $g'(c) = \frac{3}{8}$ when $c \approx 2.2$ and 6.4

c) $g'(c) = \frac{g(6) - g(2)}{6 - 2} = \frac{1 - 3}{4} = -\frac{1}{2}$, $g'(c) = -\frac{1}{2}$ when $c \approx 3.7$ and 5.5

⑤ $f(x) = 2x^2 + 4x + 5$, $[-1, 3]$. f is a polynomial hence it's cts and dt on $\mathbb{R}$, and hence $f(x)$ is cts on $[-1, 3]$ and dt on $(-1, 3)$. Since $f(-1) = f(3)$ we can use Rolle's theorem.

$$f'(c) = 4c - 4 \text{ and } f'(c) = 0 \Rightarrow c = 1.$$

$1 \in (-1, 3)$ and Rolle's theorem is satisfied.

⑧ $f(x) = x + \frac{1}{x}$, $[\frac{1}{2}, 2]$ f is continuous on its domain $(-\infty, 0) \cup (0, \infty)$ therefore also cts on $[\frac{1}{2}, 2]$

$f'(x) = \frac{x^2 - 1}{x^2}$ has the same domain and is differentiable

on $(\frac{1}{2}, 2)$. $f(\frac{1}{2}) = f(2) = \frac{5}{2}$

Now $f'(c) = \frac{c^2 - 1}{c^2} = 0 \Leftrightarrow c = \pm 1$. $1 \in (\frac{1}{2}, 2)$ and

therefore satisfies Rolle's theorem.

- (27) Does there exist a function f such that $f(0) = -1$, $f(2) = 4$ and $f'(x) \leq 2$ for all x ?
- Suppose it is possible for such a function to exist.
By the Mean Value Theorem there is a number $0 < c < 2$ such that $f'(c) = \frac{f(2) - f(0)}{2 - 0} = \frac{4 - (-1)}{2} = 5/2$.
Which is impossible since $f'(c) \leq 2$ for all x .

- (29) Show that $\sin x < x$ if $0 < x < 2\pi$

$f(x)$ is cts and dr on $\mathbb{R}$. Let $a \in (0, 2\pi)$.
Then f is cts on $[0, a]$ and f is dr on $(0, a)$.
By the mean value theorem there is a number $c \in (0, a)$ such that $f(a) - f(0) = f'(c)(a - 0)$.
So $\sin a - 0 = a \cdot \cos c < a$. As a was an arbitrary number in $(0, 2\pi)$
 $\sin x < x$ for all $0 < x < 2\pi$.

WTFW158 Unit 4.3 Exercises 4.3 p300

9-18

- (a) Find the intervals on which f is increasing or decreasing.
 (b) Find the local maximum and minimum values of f .
 (c) Find the intervals of concavity and the inflection points.

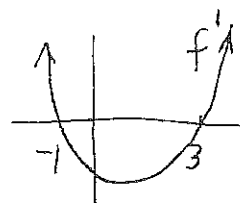
9. $f(x) = x^3 - 3x^2 - 9x + 4$

15. $f(x) = e^{2x} + e^{-x}$

16. $f(x) = x^2 \ln x$

9. $f(x) = x^3 - 3x^2 - 9x + 4$

$f'(x) = 3x^2 - 6x - 9$
 $= 3(x+1)(x-3)$



f is increasing on $(-\infty, -1)$

f is increasing on $(3, \infty)$ and f is decreasing on $(-1, 3)$

- b) f has a local max at -1 . The local max is $f(-1) = 9$
 f has a local min at 3 . The local min is $f(3) = -23$

c) $f''(x) = 6x - 6 = 6(x-1)$

$f''(x) = 0$ when $x = 1$

There is an inflection point at $(1, 7)$

$f''(x) > 0$ when $x > 1$

f is concave down on $(1, \infty)$

$f''(x) < 0$ when $x < 1$

f is concave up on $(-\infty, 1)$

15. $f(x) = e^{2x} + e^{-x}$

$f'(x) = 2e^{2x} - e^{-x}$

$f''(x) = 4e^{2x} + e^{-x}$

(a) $f'(x) = 0 \Rightarrow 2e^{2x} - e^{-x} = 0 \Rightarrow 2e^{2x} = e^{-x}$

$\Rightarrow \ln(2e^{2x}) = \ln e^{-x} \Rightarrow \ln 2 + \ln e^{2x} = \ln e^{-x}$

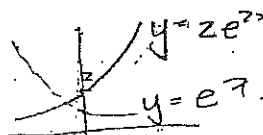
$\Rightarrow \ln 2 + 2x = -x \Rightarrow 3x = -\ln 2 \Rightarrow x = -\frac{1}{3} \ln 2$

$f'(x) > 0 \Rightarrow 2e^{2x} > e^{-x}$

$\therefore f$ is increasing on $(-\frac{1}{3} \ln 2, \infty)$

$f''(x) < 0 \Rightarrow 2e^{2x} < e^{-x}$

$\therefore f$ is decreasing on $(-\infty, -\frac{1}{3} \ln 2)$



WTW158 Unit 4.3 Exercises 4.3 p 300

(b) $\begin{array}{c} \text{decreasing} \quad \text{increasing} \\ \hline -\frac{1}{3}\ln 2 \end{array}$ f has a local minimum at $x = -\frac{1}{3}\ln 2$

the local minimum is $f(-\frac{1}{3}\ln 2) = f(\ln 2^{-\frac{1}{3}}) = e^{2(\ln 2^{-\frac{1}{3}})} + e^{-\ln 2^{-\frac{1}{3}}}$
 $= 2^{-2/3} + 2^{1/3}$

(c) $f''(x) = 0 \Rightarrow 4e^{2x} + e^{-x} = 0 \Rightarrow 4e^{2x} = -e^{-x}$
 $\Rightarrow x \in \emptyset$ (no solution as $4e^{2x} > 0$ and $-e^{-x} < 0$)

f has no inflection points

$f''(x) > 0$ on $\mathbb{R}$, f is concave upwards on $(-\infty, \infty)$

* (16) $f(x) = x^2 \ln x$

$f'(x) = 2x \ln x + x^2 \frac{1}{x} = 2x \ln x + x = x(2 \ln x + 1)$

$f''(x) = 2 \ln x + 2x \times \frac{1}{x} + 1 = 2 \ln x + 3$

(a) $f'(x) = 0 \Rightarrow x(2 \ln x + 1) = 0 \Rightarrow \underline{x = 0}$ or $2 \ln x + 1 = 0$
not in domain

$\Rightarrow \ln x = -\frac{1}{2} \Rightarrow x = e^{-\frac{1}{2}} = \frac{1}{e^{1/2}} = \frac{1}{\sqrt{e}}$

$f'(x) > 0 \Rightarrow x(2 \ln x + 1) > 0 \Rightarrow 2 \ln x + 1 > 0$ as $x > 0$

$\Rightarrow \ln x > -\frac{1}{2} \Rightarrow x > \frac{1}{\sqrt{e}}$

f is increasing on $(\frac{1}{\sqrt{e}}, \infty)$

$f'(x) < 0 \Rightarrow x(2 \ln x + 1) < 0 \Rightarrow 2 \ln x + 1 < 0$ as $x < 0$

$\Rightarrow \ln x < -\frac{1}{2} \Rightarrow x \in (0, \frac{1}{\sqrt{e}})$

f is decreasing on $(0, \frac{1}{\sqrt{e}})$

(b) $\begin{array}{c} \text{decreasing} \quad \text{increasing} \\ \hline 0 \quad \frac{1}{\sqrt{e}} \end{array}$ f has a local minimum at $x = \frac{1}{\sqrt{e}}$

The local minimum is $f(\frac{1}{\sqrt{e}}) = (\frac{1}{\sqrt{e}})^2 \ln(e^{-\frac{1}{2}}) = \frac{1}{e}(-\frac{1}{2}) \ln e$
 $= -\frac{1}{2e}$

WTW158 Unit 4.3 Exercises 4.3 p 301

$$16c) f''(x) = 0 \Rightarrow 2 \ln x + 3 = 0 \Rightarrow \ln x = -\frac{3}{2} \Rightarrow x = e^{-1.5}$$

$$f''(x) > 0 \Rightarrow \ln x > -\frac{3}{2} \Rightarrow x > e^{-1.5}$$

f is concave upwards on $(e^{-1.5}, \infty)$

$$f''(x) < 0 \Rightarrow \ln x < -\frac{3}{2} \Rightarrow x \in (0, e^{-1.5})$$

f is concave downwards on $(0, e^{-1.5})$

f has an inflection point at $x = e^{-1.5}$, point is $(e^{-3/2}, \frac{-3}{2e^{3/2}})$

30) Sketch a graph of a function that satisfies all of the given conditions.

$$f'(0) = f'(4) = 0 \quad f'(x) = 1 \text{ if } x < -1$$

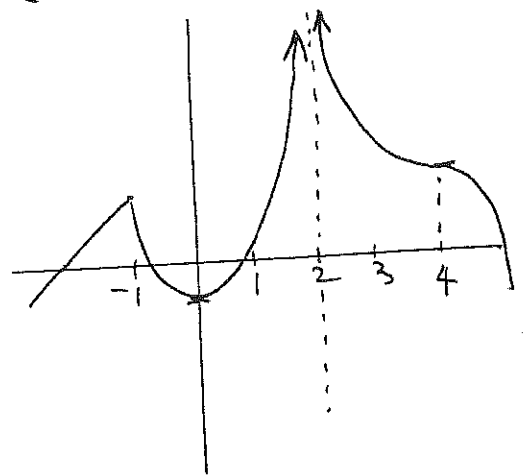
$$f'(x) > 0 \text{ if } 0 < x < 2$$

$$f'(x) < 0 \text{ if } -1 < x < 0 \text{ or } 2 < x < 4 \text{ or } x > 4$$

$$\lim_{x \rightarrow 2^-} f'(x) = \infty, \quad \lim_{x \rightarrow 2^+} f'(x) = -\infty$$

$$f''(x) > 0 \text{ if } -1 < x < 2 \text{ or } 2 < x < 4$$

$$f''(x) < 0 \text{ if } x > 4$$



* (47) $f(\theta) = 2\cos\theta + \cos^2\theta, 0 \leq \theta \leq 2\pi$

$$f'(\theta) = -2\sin\theta + 2\cos\theta(-\sin\theta) = -2\sin\theta(1+\cos\theta)$$

$$f''(\theta) = -2\cos\theta(1+\cos\theta) - 2\sin\theta(-\sin\theta)$$

$$= -2\cos\theta - 2\cos^2\theta + 2\sin^2\theta$$

$$= -2\cos\theta - 2\cos^2\theta + 2(1 - \cos^2\theta)$$

$$= -4\cos^2\theta - 2\cos\theta + 2 = -2(2\cos\theta - 1)(\cos\theta + 1)$$

(a) $f'(\theta) = 0 \Rightarrow \sin\theta = 0 \text{ or } \cos\theta = -1$

$$\Rightarrow (\theta = 0 \text{ or } \theta = \pi \text{ or } \theta = 2\pi) \text{ or } \theta = \pi$$

$$\Rightarrow \theta \in \{0, \pi, 2\pi\}$$

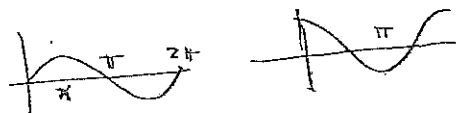
$$f''(\theta) > 0 \Rightarrow -2\sin\theta(1+\cos\theta) > 0$$

$$\Rightarrow -2\sin\theta > 0 \text{ and } 1+\cos\theta > 0 \text{ or } -2\sin\theta < 0 \text{ and } 1+\cos\theta < 0$$

$$\Rightarrow \sin\theta < 0 \text{ and } \cos\theta > -1 \text{ or } \sin\theta > 0 \text{ and } \cos\theta < -1$$

$$\Rightarrow \theta \in (\pi, 2\pi) \text{ and } \theta \in (0, 2\pi) \text{ or } \theta \in (0, \pi) \text{ and } \text{no } \theta$$

$$\Rightarrow \theta \in (\pi, 2\pi)$$



WTW158 Unit 4.3 Exercises 4.3 p 302

(4.7) f is increasing on $(\pi, 2\pi)$

$$f'(x) < 0 \Rightarrow -2\sin\theta(1+\cos\theta) < 0$$

$$\Rightarrow -2\sin\theta > 0 \text{ and } 1+\cos\theta < 0 \text{ or } -2\sin\theta < 0 \text{ and } 1+\cos\theta > 0$$

$$\Rightarrow \sin\theta < 0 \text{ and } \cos\theta < -1 \text{ or } \sin\theta > 0 \text{ and } \cos\theta > -1$$

$$\Rightarrow x \in (\pi, 2\pi) \text{ and } x \in \emptyset \text{ or } x \in (0, \pi) \text{ and } \theta \in (0, 2\pi)$$

$$\Rightarrow x \in (0, \pi)$$

$\Rightarrow f$ is decreasing on $(0, \pi)$

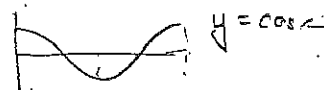
(b) $\begin{array}{c} \text{dec} \quad \text{increase} \\ \hline 0 \quad \pi \quad 2\pi \end{array}$ f has a local minimum at $x=\pi$

$$\text{The local minimum is } f(\pi) = 2\cos\pi + \cos^2\pi = -2 + 1 = -1$$

(c) $f''(\theta) = 0 \Rightarrow -2(2\cos\theta - 1)(\cos\theta + 1) = 0$

$$\Rightarrow \cos\theta = \frac{1}{2} \text{ or } \cos\theta = -1$$

$$\Rightarrow \theta = \frac{\pi}{3} \text{ or } \theta = \frac{5\pi}{3} \text{ or } \cos\theta = -1$$



$$f''(\theta) > 0 \Rightarrow -2(2\cos\theta - 1)(\cos\theta + 1) > 0$$

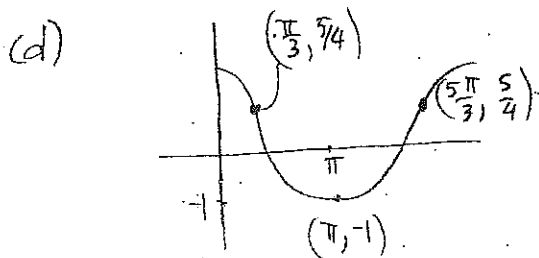
$$\Rightarrow 2\cos\theta - 1 < 0 \Rightarrow \cos\theta < \frac{1}{2} \Rightarrow \theta \in \left(\frac{\pi}{3}, \frac{5\pi}{3}\right)$$

$$f''(\theta) < 0 \Rightarrow \theta \in \left(0, \frac{\pi}{3}\right) \text{ or } \theta \in \left(\frac{5\pi}{3}, 2\pi\right)$$

f is concave upwards on $\left(\frac{\pi}{3}, \frac{5\pi}{3}\right)$

f is concave downwards on $\left(0, \frac{\pi}{3}\right)$ and $\left(\frac{5\pi}{3}, 2\pi\right)$

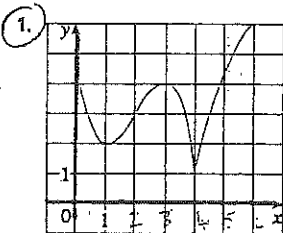
Inflection points: $\left(\frac{\pi}{3}, \frac{5}{4}\right)$ and $\left(\frac{5\pi}{3}, \frac{5}{4}\right)$



WTW158 Unit 4.3 How the derivative affects the shape of the graph.

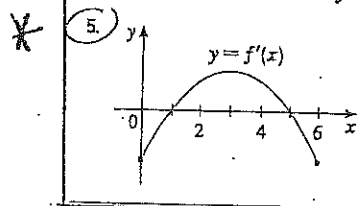
Exercises 4.3 p 300

- 1-2 Use the given graph of f to find the following.
- The open intervals on which f is increasing.
 - The open intervals on which f is decreasing.
 - The open intervals on which f is concave upward.
 - The open intervals on which f is concave downward.
 - The coordinates of the points of inflection.



- f is increasing on $(1, 3)$ and $(4, 6)$
- f is decreasing on $(0, 1)$ and $(3, 4)$
- f is concave upwards on $(0, 2)$
- f is concave downwards on $(2, 4)$ and $(4, 6)$
- The inflection point is $(2, 3)$

- 5-6 The graph of the derivative f' of a function f is shown.
- On what intervals is f increasing or decreasing?
 - At what values of x does f have a local maximum or minimum?

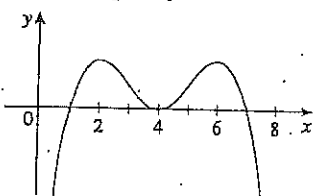


- f is increasing on $(1, 5)$ because f' is pos on $(1, 5)$
 f is decreasing on $(0, 1)$ and $(5, 6)$ because f' is neg on the interval.

- f has a local min at $x=1$ (f changes from decreasing to increasing)
 f has a local max at $x=5$ (f changes from increasing to decreasing)

7. In each part state the x -coordinates of the inflection points of f . Give reasons for your answers.

- The curve is the graph of f .
- The curve is the graph of f' .
- The curve is the graph of f'' .



f has an inflection point at the value where f changes from concave upwards to concave downwards or from concave downwards to concave upwards.

- Inflection points at $x=3$ and $x=5$

- At $x=2$ and 6 f' has a max so f'' changes from pos to neg.

at $x=4$ f' has a min so f'' changes from pos to neg

- At $x=1$ f'' changes from neg to pos so f changes from concave down to up
- Inflection points at $x=1$ and $x=7$

*74 For what values of the numbers a and b does the function

$$f(x) = ae^{bx^2}$$

have the maximum value $f(2) = 1$?

$$f(x) = ae^{bx^2}$$

$$f'(x) = ae^{bx^2} + axe^{bx^2}(2bx) = ae^{bx^2}(1 + 2bx^2)$$

$$\begin{aligned} f''(x) &= ae^{bx^2}(2bx)(1 + 2bx^2) + ae^{bx^2}4bx \\ &= 2abxe^{bx^2}(1 + 2bx^2 + 2) = 2abxe^{bx^2}(2bx + 3) \end{aligned}$$

f has a maximum value $f(2)=1$ if

$$\textcircled{1} f(2)=1 \quad \textcircled{2} f'(2)=0 \quad \textcircled{3} f''(2)<0$$

$$f(2)=1 \Rightarrow 2ae^{4b}=1 \quad \textcircled{1}$$

$$f'(2)=0 \Rightarrow ae^{4b}(1+8b)=0 \quad \textcircled{2}$$

$$f''(2)<0 \Rightarrow 4ab e^{4b}(4b+3)<0 \quad \textcircled{3}$$

$$\text{from } \textcircled{2}: 1+8b=0 \Rightarrow b=-\frac{1}{8}$$

$$\text{from } \textcircled{1}: a = \frac{1}{2e^{4b}} = \frac{1}{2e^{-1/2}} = \frac{1}{2}e^{1/2}$$

$$\text{sub in } \textcircled{3}: 2 \times \frac{1}{2} e^{-1/8} \left(-\frac{1}{8}\right) \times 2e^{-1/8(4)} \left(2 \times -\frac{1}{8} \times 2 + 3\right)$$

$$= -\frac{1}{4} \times \frac{5}{2} < 0$$

WTW 158 Unit 4.4 Indeterminate forms and L'Hospital's rule

Exercises 4.4 p 311

$$* (17) \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{1 - \sin \theta}{1 + \cos 2\theta} \left(\text{form } \frac{0}{0} \right) = \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{-\cos \theta}{-2 \sin 2\theta} \left(\text{form } \frac{0}{0} \right)$$

$$= \lim_{\theta \rightarrow \frac{\pi}{2}} \frac{\sin \theta}{-4 \cos 2\theta} = \frac{1}{-4(-1)} = \frac{1}{4}$$

$$* (19) \lim_{x \rightarrow \infty} \frac{\ln x}{\sqrt{x}} \left(\text{form } \frac{\infty}{\infty} \right) = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{\frac{1}{2}\sqrt{x}} = \lim_{x \rightarrow \infty} 2 \frac{1}{\sqrt{x}} = 0$$

$$* (25) \lim_{x \rightarrow 0} \frac{\sqrt{1+2x} - \sqrt{1-4x}}{2x} \left(\text{form } \frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\frac{1}{2} \frac{1}{\sqrt{1+2x}} \times 2 - \frac{1}{2} \frac{1}{\sqrt{1-4x}} (-4)}{1} = \lim_{x \rightarrow 0} \frac{1}{\sqrt{1+2x}} + \frac{2}{\sqrt{1-4x}} = 3$$

$$* (27) \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{2x} \left(\text{form } \frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} \left(\text{form } \frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{e^x}{2} = \frac{1}{2}$$

$$* (31) \lim_{x \rightarrow 0} \frac{\arcsin x}{x} \left(\text{form } \frac{0}{0} \right) = \lim_{x \rightarrow 0} \frac{\frac{1}{\sqrt{1-x^2}}}{1} = 1$$

$$* (55) \lim_{x \rightarrow \infty} (x - \ln x) \left(\text{form } \infty - \infty \right) = \lim_{x \rightarrow \infty} x \left(1 - \frac{\ln x}{x} \right)$$

$$\text{as } \lim_{x \rightarrow \infty} \frac{\ln x}{x} \left(\text{form } \frac{\infty}{\infty} \right) = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{1} = 0 \quad \lim_{x \rightarrow \infty} x \left(1 - \frac{\ln x}{x} \right) = \infty$$

$$* (60) \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x} \right)^{bx} \left(\text{form } 1^\infty, \text{ if } b > 0 \right) \text{ let } y = \left(1 + \frac{a}{x} \right)^{bx}$$

$$\lim_{x \rightarrow \infty} \ln y = \lim_{x \rightarrow \infty} bx \ln \left(1 + \frac{a}{x} \right) \left(\text{form } \infty \times 0 \right) = \lim_{x \rightarrow \infty} \frac{b \ln \left(1 + \frac{a}{x} \right)}{\frac{1}{x}} \left(\text{form } \frac{0}{0} \right)$$

$$= \lim_{x \rightarrow \infty} \frac{b \frac{-a/x^2}{1 + \frac{a}{x}}}{-1/x^2} = \lim_{x \rightarrow \infty} \frac{ab}{1 + \frac{a}{x}} = ab$$

$$\lim_{x \rightarrow \infty} y = \lim_{x \rightarrow \infty} e^{\ln y} = e^{ab}$$

WTW158 Unit 4.4 Exercises 4.4 p 312

$$(49) \lim_{x \rightarrow 1^+} \ln x \tan\left(\frac{\pi x}{2}\right) = \lim_{x \rightarrow 1^+} \frac{\ln x}{\cot\left(\frac{\pi x}{2}\right)} \quad (\text{form } \frac{0}{0})$$

By L'Hospital $\lim_{x \rightarrow 1^+} \frac{\frac{1}{x}}{-\frac{\pi}{2} \csc^2\left(\frac{\pi x}{2}\right)} = \frac{1}{-\frac{\pi}{2} (1)^2} = -2/\pi$

$$(57) \lim_{x \rightarrow 0^+} x^{\sqrt{x}} \quad \text{Let } y = x^{\sqrt{x}} \text{ then } \ln y = \sqrt{x} \ln x$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln x}{x^{-1/2}} \quad (\text{form } \frac{-\infty}{\infty}) \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{1/x}{-\frac{1}{2} x^{-3/2}} = -2 \lim_{x \rightarrow 0^+} \sqrt{x} = 0$$

Therefore $\lim_{x \rightarrow 0^+} x^{\sqrt{x}}$

$$= \lim_{x \rightarrow 0^+} e^{\ln y} = e^0 = 1$$

$$(58) \lim_{x \rightarrow 0^+} (\tan 2x)^x \quad \text{Let } y = (\tan 2x)^x \text{ then } \ln y = x \ln(\tan 2x)$$

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} x \ln(\tan 2x) = \lim_{x \rightarrow 0^+} \frac{\ln(\tan 2x)}{\frac{1}{x}} \quad \left(\frac{\infty}{\infty}\right)$$

By L'Hospital $\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{(\tan 2x) \cdot 2 \sec^2 2x}{-x^2}$

$$= \lim_{x \rightarrow 0^+} \frac{-2x^2 \cot 2x}{\sin 2x \cdot \cos^2 2x}$$

$$= \lim_{x \rightarrow 0^+} \frac{-2x}{\sin 2x} \cdot \lim_{x \rightarrow 0^+} \frac{x}{\cos 2x} = 1 \cdot 0 = 0$$

Now $\lim_{x \rightarrow 0^+} (\tan 2x)^x = \lim_{x \rightarrow 0^+} e^{\ln y} = e^0 = 1$

$$(65) y = (4x+1)^{\cot x} \Rightarrow \ln y = \cot x \ln(4x+1)$$

Now $\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln(4x+1)}{\tan x} \stackrel{H}{=} \lim_{x \rightarrow 0^+} \frac{\frac{4}{4x+1}}{\sec^2 x} = 4$

Therefore $\lim_{x \rightarrow 0^+} (4x+1)^{\cot x} = \lim_{x \rightarrow 0^+} e^{\ln y} = e^4$

WTW158 Unit 4.5 Curve sketching

Exercises 4.5 p 321

1. ① $f(x) = \frac{1}{5}x^5 - \frac{8}{3}x^3 + 16x$

$$f'(x) = x^4 - 8x^2 + 16 = (x^2 - 4)(x^2 - 4) = (x-2)^2(x+2)^2$$

$$f''(x) = 4x^3 - 16x = 4x(x^2 - 4) = 4x(x-2)(x+2)$$

1 Domain $\mathbb{R}$

2 Intercepts

$$f(0) = 0$$

$$f(x) = 0 \Rightarrow x \left(\frac{1}{5}x^4 - \frac{8}{3}x^2 + 16 \right) = 0$$

$$\Rightarrow x=0 \text{ or } x^2 = \frac{\frac{8}{3} \pm \sqrt{\frac{64}{9} - 4 \cdot \frac{1}{5} \cdot 16}}{\frac{2}{5}} = \frac{\frac{8}{3} \pm \sqrt{\frac{320-576}{45}}}{\frac{2}{5}} \text{ no solutions}$$
$$\Rightarrow x=0$$

3 Even/Odd?

$$f(-x) = -\frac{1}{5}x^5 + \frac{8}{3}x^3 - 16x = -\left(\frac{1}{5}x^5 - \frac{8}{3}x^3 + 16x\right) = -f(x)$$

odd

4 Horizontal asymptotes

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} x^3 \left(\frac{1}{5}x^2 - \frac{8}{3} \right) + 16x = +\infty$$

no HA,

$$\lim_{x \rightarrow -\infty} f(x) = \lim_{x \rightarrow -\infty} x^3 \left(\frac{1}{5}x^2 - \frac{8}{3} \right) + 16x = -\infty$$

5 Vertical asymptotes

Domain = $\mathbb{R}$ no VA

6. Slant asymptotes

None

7. $f'(x) > 0 \Rightarrow (x-2)^2(x+2)^2 > 0 \Rightarrow x \neq 2 \text{ and } x \neq -2$

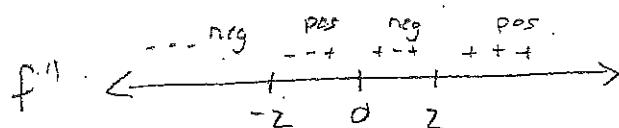
f is increasing on $(-\infty, -2)$, $(-2, 2)$ and $(2, \infty)$

7. 8 Local extremes

None as f is always increasing

9 Concave up/down

$$f''(x) = 4x(x-2)(x+2)$$



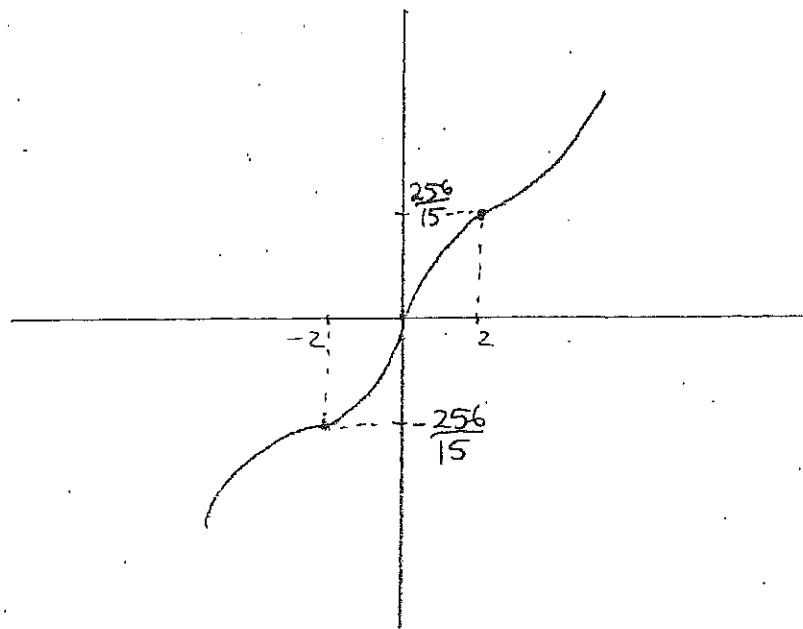
f is concave down ward on $(-\infty, -2)$ and $(0, 2)$

f is concave upward on $(-2, 0)$ and $(2, \infty)$

10 Inflection points

$$f(-2) = -17.06, \quad f(0) = 0, \quad f(2) = 17.06 \text{ are inflection points}$$

Sketch.



WTW158. Unit 4.5 Exercises 4.5 p 322

*27 $y = f(x) = \frac{\sqrt{1-x^2}}{x}$

$$f'(x) = \frac{\frac{1}{2}(1-x^2)^{-\frac{1}{2}}(-2x) \cdot x - \sqrt{1-x^2} \cdot 1}{x^2} = \frac{-x^2}{\sqrt{1-x^2} \cdot x^2} - \frac{\sqrt{1-x^2}}{x^2}$$

$$= \frac{-x^2 - (1-x^2)}{x^2 \sqrt{1-x^2}} = \frac{-1}{x^2 \sqrt{1-x^2}}$$

$$f''(x) = \frac{d}{dx} -x^{-2}(1-x^2)^{-\frac{1}{2}} = (-1)(-2)x^{-3}(1-x^2)^{-\frac{1}{2}} + (-x^{-2})\left(\frac{-1}{2}\right)(1-x^2)^{-\frac{3}{2}}(-2x)$$

$$= \frac{2}{x^3 \sqrt{1-x^2}} + \frac{-1}{x(1-x^2)^{3/2}} = \frac{2(1-x^2) + x^2}{x^3(1-x^2)^{3/2}}$$

$$= \frac{2-3x^2}{x^3(1-x^2)^{3/2}}$$

1. Domain

$$D = \{x \mid 1-x^2 \geq 0 \text{ and } x \neq 0\} = \{x \mid (1-x)(1+x) \geq 0 \text{ and } x \neq 0\}$$

$$= [-1, 0) \cup (0, 1]$$

2. Intercepts

$f(0)$ is undefined $\Rightarrow$ no y-intercept

$$f(x) = 0 \Rightarrow 1-x^2 = 0 \Rightarrow x = 1 \text{ or } x = -1$$

3. Even or odd?

$$f(-x) = \frac{\sqrt{1-(-x)^2}}{-x} = \frac{\sqrt{1-x^2}}{-x} = -\frac{\sqrt{1-x^2}}{x} = -f(x)$$

f is odd.

4. Horizontal asymptotes

None as domain $= [-1, 0) \cup (0, 1]$

27) 5 Vertical asymptotes

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\sqrt{1-x^2}}{x} = +\infty$$

$\therefore x=0$ is a VA

$$\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} \frac{\sqrt{1-x^2}}{x} = -\infty$$

6 Slant asymptotes

None

7 Increasing and decreasing

$$f'(x) = \frac{-1}{\sqrt{1-x^2} x^2} < 0 \text{ if } x \neq 1, -1, 0$$

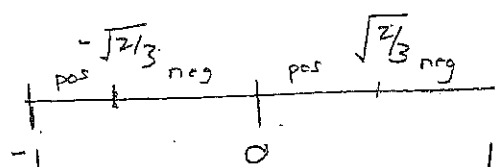
f is decreasing on $(-1, 0)$ and on $(0, 1)$.

8 Local extremes

None

9 Concavity

$$f''(x) = \frac{2-3x^2}{x^3(1-x^2)^{3/2}}$$



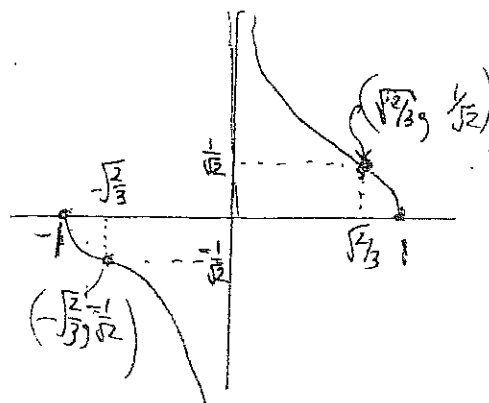
$$= \frac{(\sqrt{2}-\sqrt{3}x)(\sqrt{2}+\sqrt{3}x)}{x^3(1-x^2)^{3/2}}$$

f is concave up on $(-1, -\sqrt{2}/3)$ and on $(0, \sqrt{2}/3)$

f is concave downward on $(-\sqrt{2}/3, 0)$ and on $(\sqrt{2}/3, 1)$

10. Inflection points

$$(\sqrt{2}/3, \frac{1}{\sqrt{2}}), (-\sqrt{2}/3, -\frac{1}{\sqrt{2}})$$



WTW158 Unit 4.5 Exercises 4.5 p 322

* 52 $y = f(x) = \frac{\ln x}{x^2}$

$$f'(x) = \frac{\frac{1}{x} \cdot x^2 - \ln x \cdot 2x}{x^4} = \frac{x - 2x \ln x}{x^4} = \frac{x(1 - 2 \ln x)}{x^4}$$

$$= \frac{1 - 2 \ln x}{x^3}$$

$$f''(x) = \frac{\left(-\frac{2}{x}\right)x^3 - (1 - 2 \ln x) \cdot 3x^2}{x^6} = \frac{-2x^2 - 3x^2 + 6x^2 \ln x}{x^6}$$

$$= \frac{-5x^2 + 6x^2 \ln x}{x^6} = \frac{x^2(-5 + 6 \ln x)}{x^6} = \frac{-5 + 6 \ln x}{x^4}$$

1. Domain

$(0, \infty)$ because $\ln x$ is only defined if $x > 0$

2. Intercepts

$f(0)$ is undefined $\Rightarrow$ no y-intercept

$f(x) = 0 \Rightarrow \ln x = 0 \Rightarrow x = 1$

3. Even or odd

Neither, as the domain is $(0, \infty)$

4. Horizontal Asymptotes

$\lim_{x \rightarrow -\infty} f(x)$ is undefined as domain $= (0, \infty)$

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{\ln x}{x^2} = \lim_{x \rightarrow \infty} \frac{\frac{1}{x}}{2x} = \lim_{x \rightarrow \infty} \frac{1}{2x^2} = 0 \quad \text{Using L'Hôpital's}$$

$\therefore y = 0$ is a HA

5) Vertical asymptotes

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} \frac{\ln x}{x^2} = -\infty \quad \therefore x = 0 \text{ is a VA}$$

6. Slant asymptote

None

7. Increasing/decreasing

$$f'(x) = \frac{1-2\ln x}{x^2}$$

$$f'(x) = 0 \Rightarrow 1-2\ln x = 0 \Rightarrow \ln x = \frac{1}{2} \Rightarrow x = e^{\frac{1}{2}} = \sqrt{e}$$

$$f'(x) > 0 \Rightarrow 1-2\ln x > 0 \Rightarrow \ln x < \frac{1}{2} \Rightarrow x < e^{\frac{1}{2}} = \sqrt{e}$$

f is increasing on $(0, \sqrt{e})$

f is decreasing on $(\sqrt{e}, \infty)$

8. Local extremes

f has a local maximum at $x = \sqrt{e}$

$$f(\sqrt{e}) = \frac{\ln \sqrt{e}}{(\sqrt{e})^2} = \frac{\frac{1}{2}}{e} = \frac{1}{2e}$$

9. Concavity

$$f''(x) = \frac{-5+6\ln x}{x^4}$$

$$f''(x) > 0 \Rightarrow -5+6\ln x > 0 \Rightarrow 6\ln x > 5 \Rightarrow \ln x > \frac{5}{6} \Rightarrow x > e^{5/6}$$

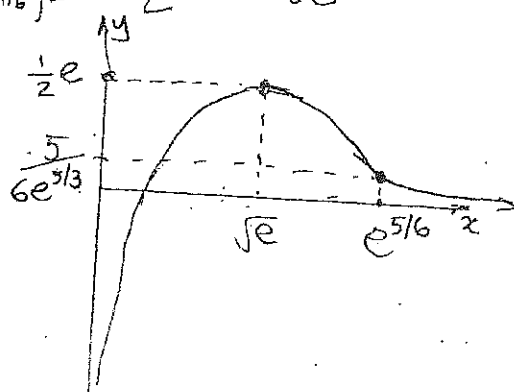
f is concave upwards on $(e^{5/6}, \infty)$

f is concave downwards on $(0, e^{5/6})$

10. Inflection points

f has an inflection point at $x = e^{5/6}$

$$f(e^{5/6}) = \frac{\ln e^{5/6}}{(e^{5/6})^2} = \frac{5/6}{e^{5/3}} = \frac{5}{6e^{5/3}}$$



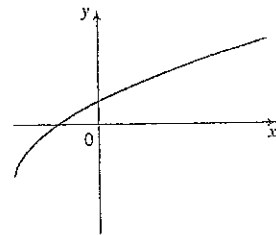
WTW158 Unit 4.5 Ex 4.3 p301

25. (a) $f'(x) > 0$ and $f''(x) < 0$ for all x
 (b) $f'(x) < 0$ and $f''(x) > 0$ for all x

26. $f'(1) = f'(-1) = 0$, $f'(x) < 0$ if $|x| < 1$,
 $f'(x) > 0$ if $1 < |x| < 2$, $f'(x) = -1$ if $|x| > 2$,
 $f''(x) < 0$ if $-2 < x < 0$, inflection point $(0, 1)$

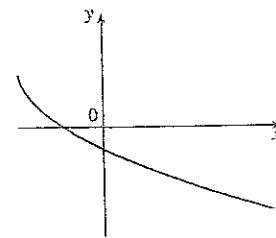
25. (a) $f'(x) > 0$ and $f''(x) < 0$ for all x

The function must be always increasing (since the first derivative is always positive) and concave downward (since the second derivative is always negative).



- (b) $f'(x) < 0$ and $f''(x) > 0$ for all x

The function must be always decreasing (since the first derivative is always negative) and concave upward (since the second derivative is always positive).

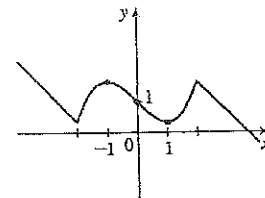


26. $f'(1) = f'(-1) = 0 \Rightarrow$ horizontal tangents at $x = \pm 1$.

$f'(x) < 0$ if $|x| < 1 \Rightarrow$ f is decreasing on $(-1, 1)$.

$f'(x) > 0$ if $1 < |x| < 2 \Rightarrow$ f is increasing on $(-2, -1)$ and $(1, 2)$.

$f'(x) = -1$ if $|x| > 2 \Rightarrow$ the graph of f has constant slope -1 on $(-\infty, -2)$ and $(2, \infty)$.



$f''(x) < 0$ if $-2 < x < 0 \Rightarrow$ f is concave downward on $(-2, 0)$. The point $(0, 1)$ is an inflection point.

27. $f'(0) = f'(2) = f'(4) = 0$,

$f'(x) > 0$ if $x < 0$ or $2 < x < 4$,

$f'(x) < 0$ if $0 < x < 2$ or $x > 4$,

$f''(x) > 0$ if $1 < x < 3$, $f''(x) < 0$ if $x < 1$ or $x > 3$

28. $f'(x) > 0$ for all $x \neq 1$, vertical asymptote $x = 1$,

$f''(x) > 0$ if $x < 1$ or $x > 3$, $f''(x) < 0$ if $1 < x < 3$

27. $f'(0) = f'(2) = f'(4) = 0 \Rightarrow$ horizontal tangents at $x = 0, 2, 4$.

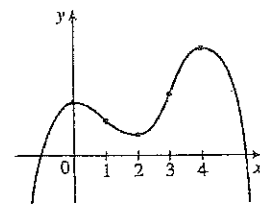
$f'(x) > 0$ if $x < 0$ or $2 < x < 4 \Rightarrow$ f is increasing on $(-\infty, 0)$ and $(2, 4)$.

$f'(x) < 0$ if $0 < x < 2$ or $x > 4 \Rightarrow$ f is decreasing on $(0, 2)$ and $(4, \infty)$.

$f''(x) > 0$ if $1 < x < 3 \Rightarrow$ f is concave upward on $(1, 3)$.

$f''(x) < 0$ if $x < 1$ or $x > 3 \Rightarrow$ f is concave downward on $(-\infty, 1)$

and $(3, \infty)$. There are inflection points when $x = 1$ and 3 .



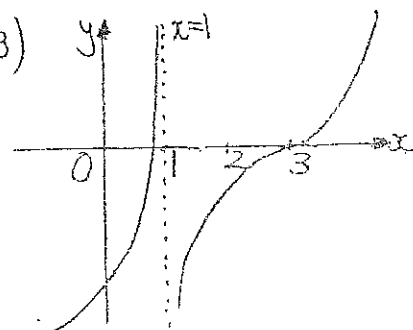
28. $f'(x) > 0$ for all $x \neq 1 \Rightarrow$ f is increasing on $(-\infty, 1) \cup (1, \infty)$

Vertical asymptote $x = 1$

$f''(x) > 0$ if $x < 1$ or $x > 3 \Rightarrow$ f concave up on $(-\infty, 1) \cup (3, \infty)$

$f''(x) < 0$ if $x \in (1, 3) \Rightarrow$ f concave down on $(1, 3)$

There is an inflection point at $x = 3$



WTW158 Optimization problems

Unit 4.6 Exercises 4.7 p337

1. Consider the following problem: Find two numbers whose sum is 23 and whose product is a maximum.

- (a) Make a table of values, like the following one, so that the sum of the numbers in the first two columns is always 23. On the basis of the evidence in your table, estimate the answer to the problem.

First number	Second number	Product
1	22	22
2	21	42
3	20	60
...	...	...

- (b) Use calculus to solve the problem and compare with your answer in a

number 1	number 2	product
1	22	22
2	21	42
3	20	60
4	19	76
5	18	90
6	17	102
7	16	112
8	15	120
9	14	126
10	13	130
11	12	132
12	11	132
13	10	130

max
some
here

- (b) let the first number be a and the second be b .
then $a+b=23$ and ab must be a maximum.

$$f(a) = ab = a(23-a) = 23a - a^2$$

$$f'(a) = 23 - 2a \text{ and } f''(a) = -2$$

$$f'(a) = 0 \Rightarrow a = \frac{23}{2} = 11.5$$

$$f''(11.5) = -2 < 0$$

f has a maximum if $a = 11.5$ and $b = 23 - a = 23 - 11.5 = 11.5$

*

3. Find two positive numbers whose product is 100 and whose sum is a minimum.

Let a and b be the numbers.

Then $a > 0$, $b > 0$, $ab = 100$, and $a+b$ must be a minimum

$$f(a) = a+b = a + \frac{100}{a} = a + 100a^{-1}$$

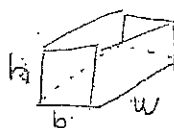
$$f'(a) = 1 - 100a^{-2} \quad f''(a) = 200a^{-3}$$

$$f'(a) = 0 \Rightarrow 1 - \frac{100}{a^2} = 0 \Rightarrow 1 = \frac{100}{a^2} \Rightarrow a^2 = 100 \Rightarrow a = 10 \quad (a \text{ is pos})$$

$$f''(10) = \frac{200}{1000} > 0$$

f has a minimum if $a = 10$ and $b = \frac{100}{a} = \frac{100}{10} = 10$

5. A rectangular storage container with an open top is to have a volume of 10 m^3 . The length of its base is twice the width. Material for the base costs \$10 per square meter. Material for the sides costs \$6 per square meter. Find the cost of materials for the cheapest such container.



given $h \times b \times w = 10$, $w = 2b$

$$\text{cost} = bw \times 10 + 2bh \times 6 + 2wh \times 6$$

$$h = \frac{10}{bw} = \frac{10}{b \cdot 2b} = \frac{10}{2b^2} = \frac{5}{b^2}$$

$$\begin{aligned} \text{cost} &= 10 \times b \times 2b + 12 \times b \times \frac{5}{b^2} + 12 \times 2b \times \frac{5}{b^2} \\ &= 20b^2 + \frac{60}{b} + \frac{120}{b} = 20b^2 + \frac{180}{b} \end{aligned}$$

$$f(b) = 20b^2 + \frac{180}{b}$$

$$f'(b) = 40b - \frac{180}{b^2} \quad f''(b) = 40 + \frac{2 \times 180}{b^3}$$

$$\begin{aligned} f'(b) = 0 &\Rightarrow 40b - \frac{180}{b^2} = 0 \Rightarrow 40b = \frac{180}{b^2} \Rightarrow b^3 = \frac{180}{40} = \frac{9}{2} \\ &\Rightarrow b = \sqrt[3]{\frac{9}{2}} \end{aligned}$$

$$f''(\sqrt[3]{\frac{9}{2}}) > 0$$

$$f \text{ has a min if } b = \sqrt[3]{\frac{9}{2}}$$

The minimum cost is ≈ 163.54